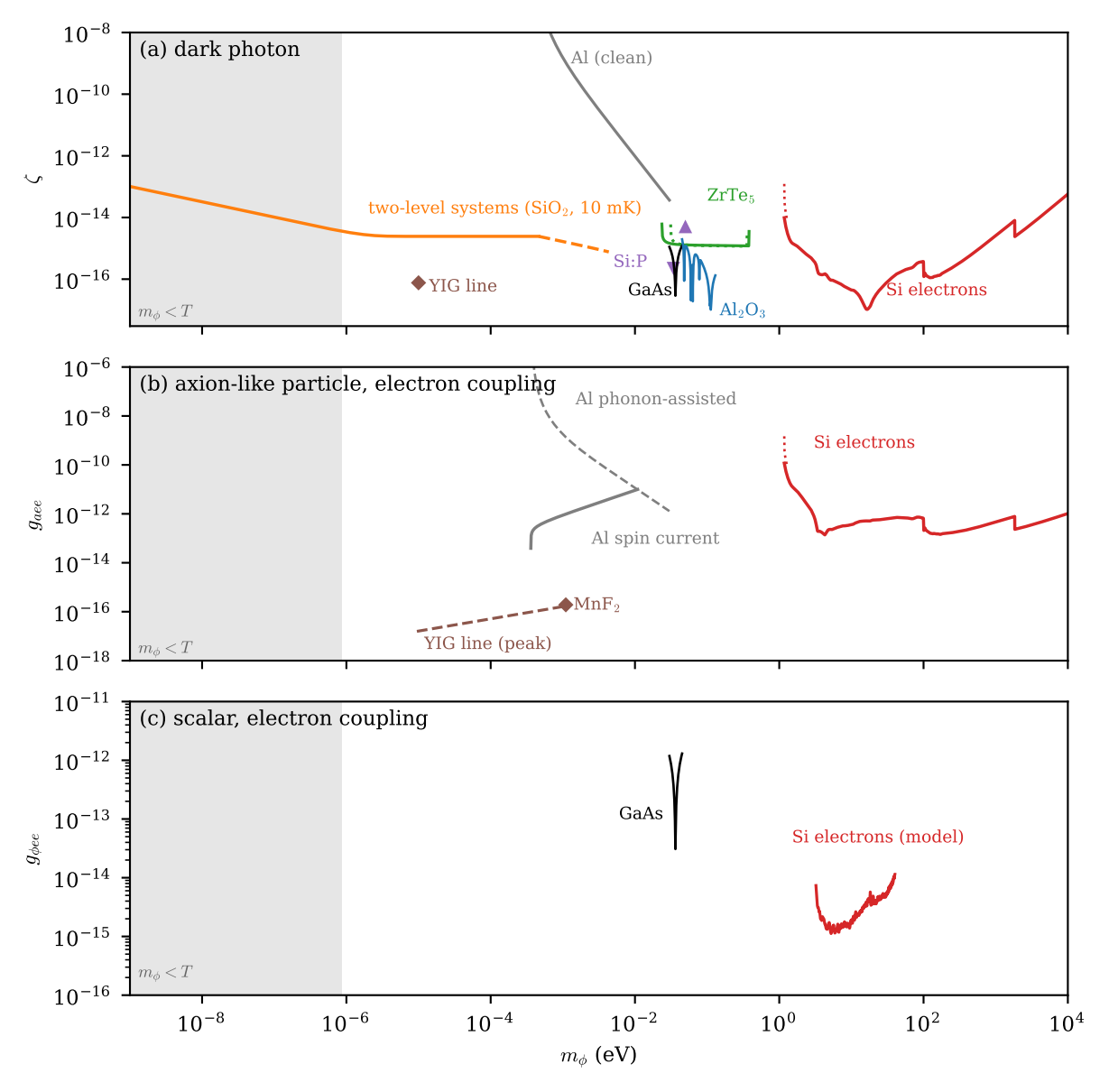
(a) dark photon
Al (clean)
two-level systems (SiO_2, 10 mK)
$ZrTe_5$
Si:P
GaAs
Al_2O_3
YIG line
Si electrons
$m_\phi < T$
ζ
(b) axion-like particle, electron coupling
Al phonon-assisted
Si electrons
Al spin current
MnF_2
YIG line (peak)
$m_\phi < T$
g_{aee}
(c) scalar, electron coupling
GaAs
Si electrons (model)
$m_\phi < T$
$g_{\phi ee}$
m_ϕ (eV)